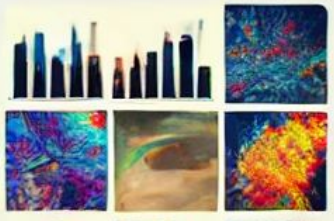
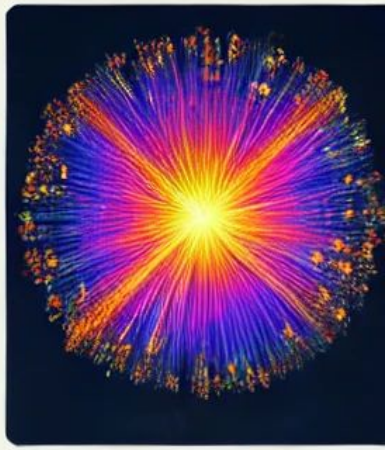
# Unsupervised Learning : Clustering

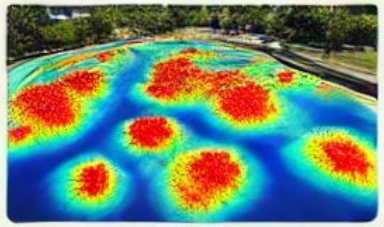
Clustering is an unsupervised learning technique that groups similar data points together based on their characteristics.

K-Means Clustering: partitions data into K clusters by minimizing distances within each cluster.



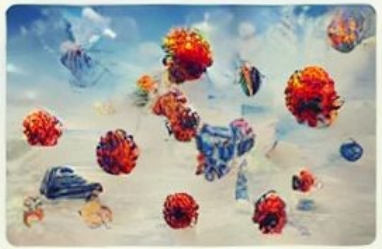



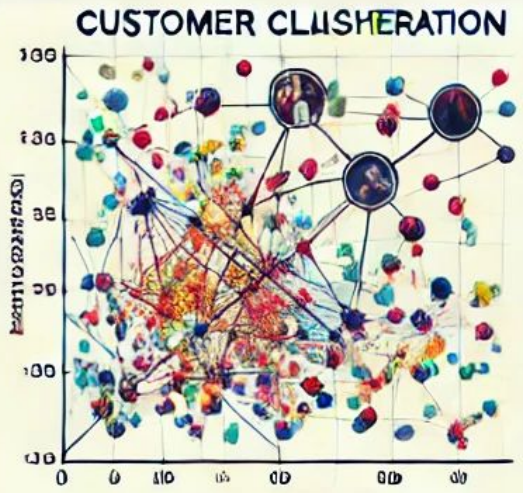
CUSTOMER
SEGMENTATION


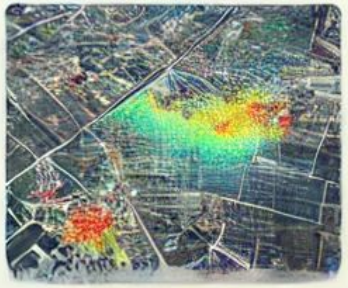



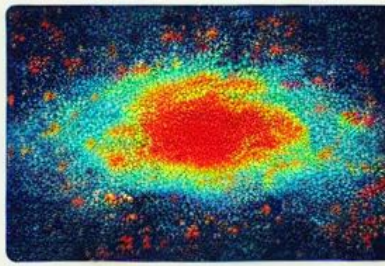
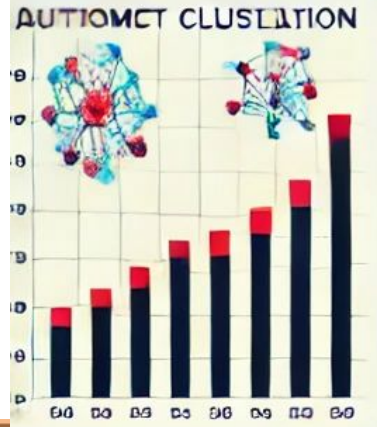
CUSTOMER SEGMENTATION

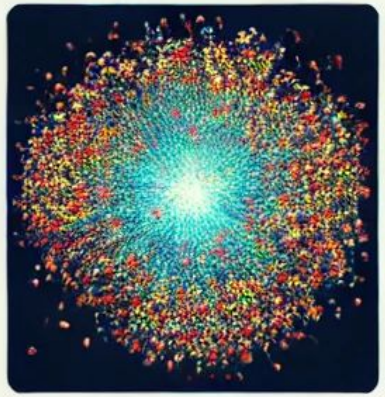


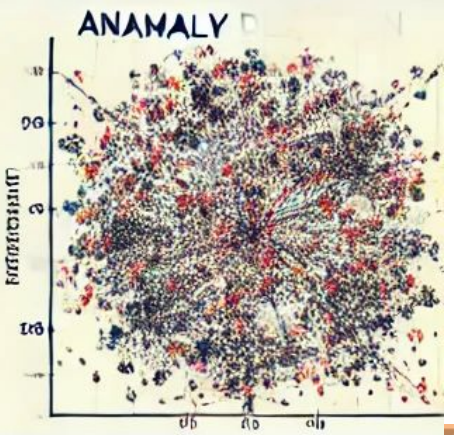



CUSTOMER CLUSTERATION

CUSTOMER DETECT
SEGMENTATION


AUTOMCT CLUSTATION

CUSTOMER DETECTION



SOCIAL NETWORK DETECTION

ANAMALY

# K-Means Clustering - A Simple Example

- **How It Works**:
  1. Choose the number of clusters k.
  2. Initialize kkk random centroids.
  3. Assign each data point to the nearest centroid.
  4. Update centroids based on assigned points.
  5. Repeat steps 3-4 until centroids stabilize.

# Example

A retail store wants to segment its customers into different groups based on two factors: **Annual Income** and **Spending Score** (a measure of spending behavior on a scale of 1 to 100). The store aims to understand customer patterns to improve targeted marketing. They want to use **K-Means clustering** to group customers into 3 distinct clusters.

| Customer ID | Annual Income ($) | Spending Score (1-10 |
|---|---|---|
| 1 | 15000 | 39 |
| 2 | 16500 | 81 |
| 3 | 18000 | 6 |
| 4 | 19000 | 77 |
| 5 | 20500 | 40 |
| 6 | 22000 | 76 |
| 7 | 23500 | 6 |
| 8 | 25000 | 94 |
| 9 | 26500 | 3 |
| 10 | 28000 | 72 |

# Step 1:

Select Random Centroid:

- Randomly select 3 initial centroids.

  Let’s take the first three customers as the initial centroids:

  - **Centroid 1**: (15,000, 39)
  - **Centroid 2**: (16,500, 81)
  - **Centroid 3**: (18,000, 6)

# Step 2:

**Assign Each Customer to the Nearest Centroid**

Calculate the Euclidean distance between each customer and each centroid.

With customer 1 & centroid 1 = 0

With customer 1 & centroid 2 =1571.34

$$\sqrt{(15000 - 16500)^2 + (39 - 81)^2} \approx 1571.34$$

# Step 3:

**Update Centroids**

After all customers are assigned to clusters, update each centroid by calculating the mean **Income** and **Spending Score** of customers in each cluster.

# Step 4

**Repeat Steps 3-4 Until Convergence**

- Reassign customers based on the new centroids and update the centroids. Continue until the centroids do not change significantly.

# Step 5

**Analyze Clusters**

- Once the algorithm converges, interpret each cluster to understand patterns. For example:
  - **Cluster 1**: Low income, moderate spending
  - **Cluster 2**: Moderate income, high spending
  - **Cluster 3**: High income, low spending

# To Do

A digital marketing team is analyzing website traffic and wants to segment visitors based on **Time Spent on Site** and **Pages Viewed** to understand different engagement levels. Using K-Means clustering, create 3 clusters of visitors. What kind of visitor types could each cluster represent?

# Data

| Visitor ID | Time Spent on Site (r | Pages Viewed |
|---|---|---|
| 1 | 2 | 1 |
| 2 | 5 | 2 |
| 3 | 15 | 5 |
| 4 | 30 | 7 |
| 5 | 5 | 2 |
| 6 | 3 | 1 |
| 7 | 45 | 10 |
| 8 | 20 | 5 |
| 9 | 10 | 3 |
| 10 | 25 | 6 |